AI Hackathon- AI-Based Student Profiling & Dropout Prevention

Pre-Orientation Briefing Document

For Participating Startups & Technology Partners

01 PROBLEM OVERVIEW

Andhra Pradesh operates one of India's largest school networks, serving students across urban, semi-urban, and rural geographies. Despite sustained investment in infrastructure and welfare schemes, the state continues to face high dropout rates — particularly among students from socio-economically disadvantaged backgrounds, rural communities, and first-generation learners.

The core challenge is the absence of a real-time, data-driven system that can identify at-risk students before they disengage or drop out. Traditional methods — teacher observation, periodic reports, manual record-keeping — are reactive rather than preventive. By the time a student is identified as at risk, it is often too late to intervene effectively.

Root Causes of Dropout

- Poverty and household financial pressure
- Early marriage (particularly among girls)
- Child labour — agriculture and informal sectors
- Poor learning outcomes and rote instruction
- Lack of personalised academic support
- Distance to school / poor infrastructure
- Seasonal migration of farming families
- Language and cultural barriers in tribal areas
- Disability without adequate school support
- Low parental literacy and engagement
- Mental health challenges — stress, disengagement
- COVID-19 learning loss compounding gaps

⚠ Problem Statement for Hackathon Participants

Design and build an AI-powered student profiling and early warning system that ingests multi-source data (academic performance, attendance, socio-economic background, behavioural indicators) to predict student dropout risk, enable proactive personalised intervention, and provide actionable dashboards for teachers, school administrators, and government officials in real time.

02 DATA LANDSCAPE & AI OPPORTUNITY

Available & Recommended Data Sources

Data Category	Availability	Key Fields	Source
Academic Records	Will be provided	Exam scores, grade progression, subject-wise marks, repetition history	APSCERT / School MIS
Attendance Data	Will be provided	Daily attendance, streak of absences, seasonal absence patterns	LEAP, School registers
Socio-Economic Profile	Assumed dataset	Family income group, BPL status, caste category, welfare scheme linkage	Assumed / synthetic
Household Indicators	Assumed dataset	Parent education level, occupation type, number of siblings, migration history	Assumed / synthetic
Behavioural Signals	Assumed dataset	Class participation, disciplinary flags, peer interaction, self-reported motivation	Assumed / synthetic

Geographic Data	Will be provided	Distance from school, transport availability, rural/urban/tribal classification	GIS / Revenue maps

Data Availability Note

- Attendance & Academic Records — will be furnished by RTGS from school MIS / UDISE+.
- Socio-Economic, Household & Behavioural data — participants must design and assume their own representative dataset for the prototype; guidance on schema will be provided.
- Geographic data — will be provided in GIS-compatible format by RTGS.

AI Techniques Applicable to This Problem

Predictive Modelling

- Random Forest & XGBoost for dropout risk scoring
- LSTM / RNN for temporal attendance pattern analysis
- Logistic Regression as interpretable baseline model
- Hybrid Logistic-Neural models for high accuracy

Natural Language Processing

- Sentiment analysis on teacher feedback notes
- Multilingual NLP for Telugu/Urdu student records
- GenAI for personalised intervention message drafting

Explainable AI (XAI)

- SHAP values to explain per-student risk factors
- LIME for local model-agnostic explanations
- Interpretable dashboards for non-technical teachers

Adaptive & Personalisation Engines

- Recommendation engines for targeted remedial support
- Adaptive learning path generation per student profile
- NPS-based longitudinal sentiment tracking

Andhra Pradesh Context — What Has Already Been Done

- AP Government partnered with Microsoft Azure ML to cover a large number of students across government schools — early dropout prediction using ML on complex datasets.
- Challenge: Existing systems remain siloed, reactive, and lack real-time student profiling that feeds into actionable intervention workflows.

03 EXPECTED SOLUTION ARCHITECTURE

Participating startups are expected to design a comprehensive, end-to-end AI system. Solutions do not need to solve every layer at hackathon stage but must clearly articulate the full architecture and deliver a working proof of concept for at least the core prediction and alert layers.

Layer	Component	What It Does
L1	Data Ingestion & Integration	Connects UDISE+, school MIS, welfare databases; handles missing data, normalises records across schools
L2	Student Profiling Engine	Creates a holistic risk profile per student: academic trajectory, attendance trend, socio-economic score, behavioural flags
L3	Dropout Risk Prediction Model	ML model (ensemble preferred) outputs a risk score (0–100) and tier: Low / Medium / High / Critical
L4	Explainability Layer	SHAP/LIME explanations surfaced per student — tells teachers WHY a student is flagged.

L5	Intervention Recommendation	AI-generated personalised action plan: counselling, scholarship alerts, peer mentoring, remedial classes, parent SMS
L6	Dashboard & Alerting	Role-based dashboards for teacher, HM, district officer, RTGS — real-time alerts, trends, and drill-down capability
L7	Feedback Loop	Post-intervention outcome tracking: did the intervention work? Feeds back to retrain the model each semester.

04 SOLUTION REQUIREMENTS & EVALUATION CRITERIA

Mandatory Requirements (Must-Have)

1. AI/ML-based dropout risk scoring with a clearly defined risk tier (Low / Medium / High / Critical) per student.
2. Multi-dimensional student profiling incorporating at least 3 data categories (academic, attendance, socio-economic minimum).
3. Explainability: The system must explain in plain language why a student is at risk — teachers cannot act on a black-box score.
4. Role-based dashboard: Distinct views for classroom teacher, headmaster, and district/state administrator.
5. Intervention recommendation module: Auto-generate personalised alerts and suggested actions per flagged student.
6. Scalability consideration: Solution must function across large school networks; single-school demos are not sufficient.
7. Data privacy and ethics: Sensitive student data handling — anonymisation, consent logic, and PDPB 2023 alignment.

Desirable Features (Value-Add)

- Offline / low-bandwidth mode for rural schools
- Telugu language interface for teachers
- Parent engagement module via SMS/WhatsApp
- Integration with AP welfare scholarship systems
- AI-generated weekly student progress summaries
- Peer comparison and cohort-level analytics
- Longitudinal tracking across academic years
- Automated reduction of teacher reporting workload

Evaluation Criteria

Criterion	What Judges Look For
Technical Soundness of AI Model	Accuracy, recall, precision; choice of algorithm justified with rationale
Practical Usability & UX	Can a village schoolteacher use this in under 2 minutes?
Depth of Student Profiling	Breadth of data signals used; robustness of profiling logic
Intervention Design & Impact	Quality and personalisation of recommended interventions
Explainability & Ethics	Can the system explain its decisions? Are privacy safeguards in place?
Scalability & Deployment Plan	Realistic pathway to state-wide deployment within 12 months

05 INNOVATION FOCUS AREAS

RTGS and RTIH are especially interested in innovation beyond the baseline prediction model. Teams that address any of these focus areas will receive special recognition:

Focus Area 1	Focus Area 2	Focus Area 3
Hyper-Early Detection	**GenAI Counsellor Assist**	**Closed-Loop Feedback**
Identify at-risk students within the first 4–8 weeks of an academic year using pre-entry data and early behaviour signals — before grades or attendance data is even substantial.	A GenAI-powered assistant for school counsellors that reads a student's risk profile and generates personalised conversation guides, parent communication scripts, and remediation plans.	A self-improving system that captures intervention outcomes and uses them to continuously retrain the dropout model — ensuring the AI gets smarter every semester.

06 ETHICAL GUIDELINES & CONSTRAINTS

AI-based student profiling carries significant ethical responsibility. All solutions submitted must address the following non-negotiable ethical considerations:

Privacy & Data Protection

- No personally identifiable data exposed in dashboards without need-to-know access controls
- All student data must be anonymised at the analytics layer
- Compliance with PDPB 2023 (Digital Personal Data Protection Act)
- No third-party data sale or external API calls with student PII

Algorithmic Fairness

- Models must be tested for bias across gender, caste, religion, and geography
- Disproportionate flagging of marginalised groups must be audited and corrected
- Risk scores must never be used for punitive school policies

Transparency & Human Oversight

- All AI decisions must be explainable in Telugu and English to the responsible teacher
- Human teacher/counsellor must remain the final decision-maker — AI supports, never overrides
- Audit logs must be maintained for all automated interventions

Student Dignity

- Risk scores must never be shared directly with students or parents without professional mediation
- Labels such as 'dropout risk' must never appear on student-facing interfaces
- Interventions must be framed positively — support-focused, not stigmatising